

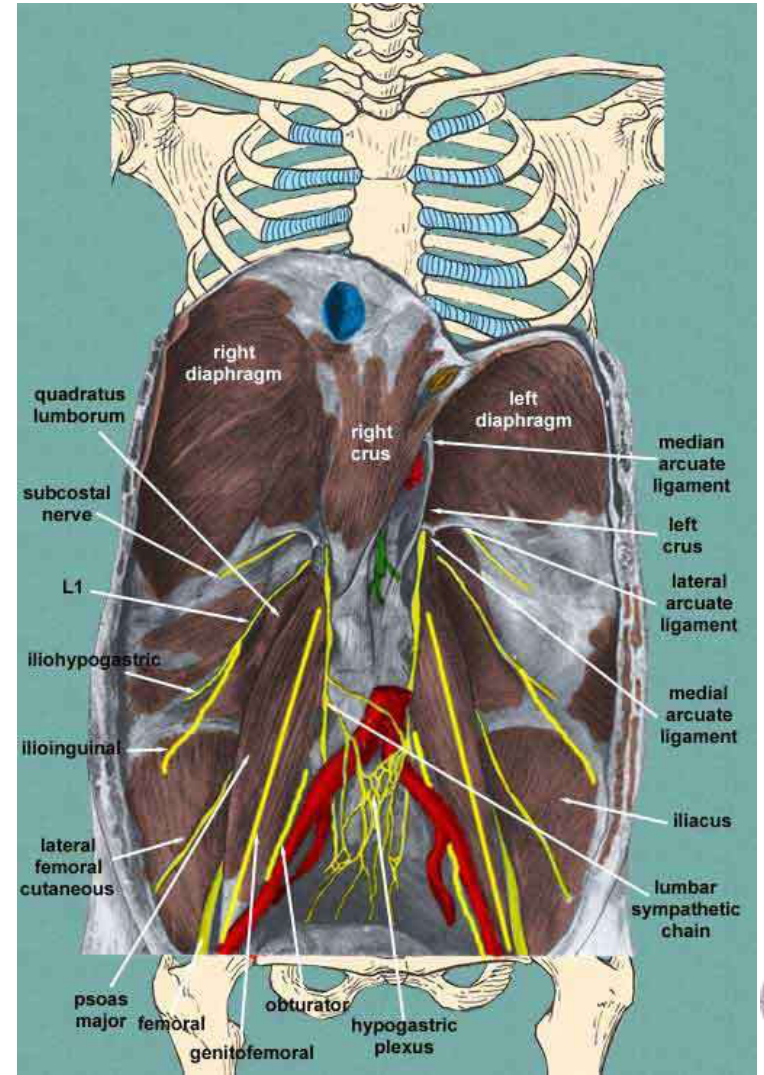
Nerves of PAW

BY: M. G. HAMMAD



Nerves of PAW

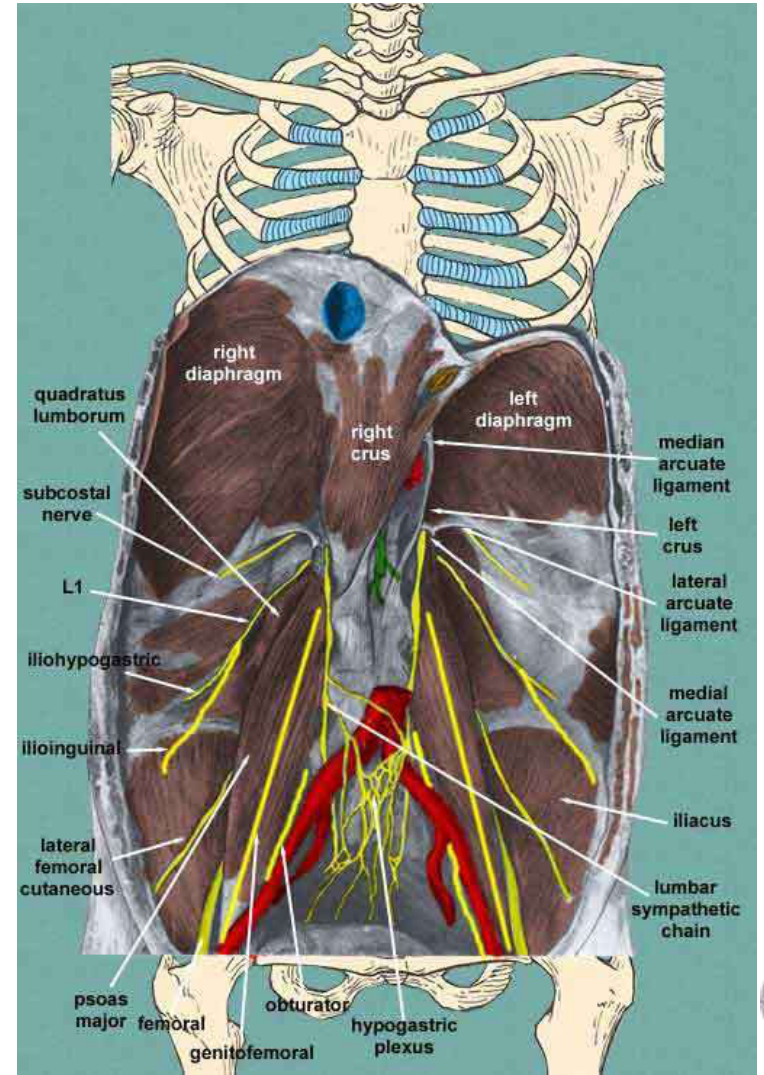
- Include:
 - ❑ Subcostal N.
 - ❑ Lumbar plexus.
 - ❑ ANS (Sympathetic & parasympathetic)



Nerves of PAW

- **Subcostal Nerve T12:**

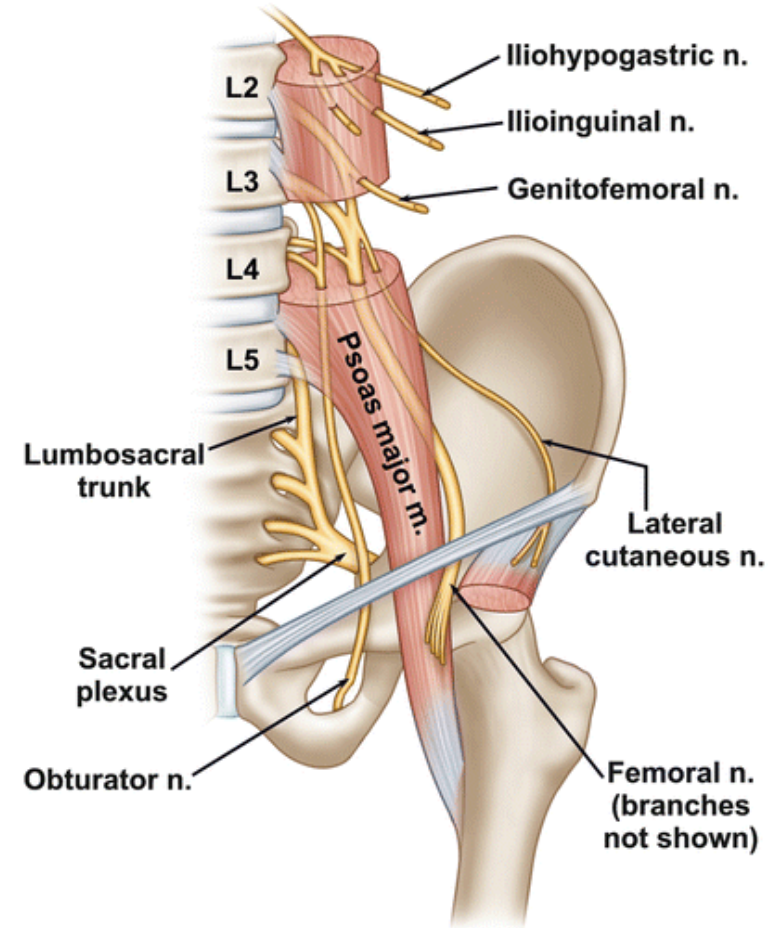
- ❑ Passes from the thorax behind the lateral arcuate ligament.
- ❑ It runs behind the kidney.
- ❑ It supplies:
 - AAW muscles.
 - Skin over them.



Nerves of PAW

lumbar plexus

- ❑ Lies in the posterior part of the substance of the psoas major muscle.
- ❑ Formation:
 - Anterior primary rami of L1-L4.
 - Enter the substance of psoas.
 - Give off branches to psoas and quadratus.
 - After this they form a plexus (LUMBAR PLEXUS) within substance of psoas major.



Lumbar plexus

- **Branches:**

- **Before divisions:**

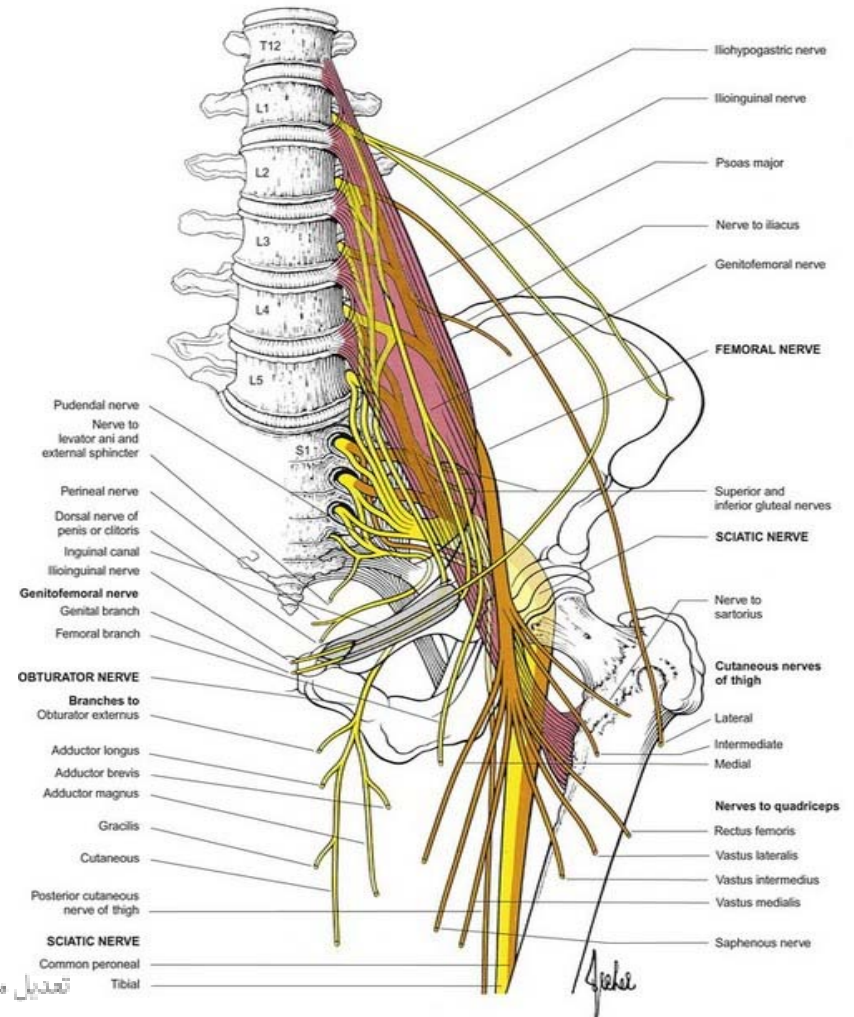
- Iliohypogastric & ilioinguinal nerves: L1.
- Genitofemoral nerve: L1, L2.

- **Anterior division:**

- Obturator nerve: L2, L3, L4.
- Lumbosacral trunk: L4, L5.

- **Posterior division:**

- Femoral nerve: L2, L3, L4.
- Lateral cutaneous branch of thigh: L2, L3.



Lumbar plexus

Nerve	Segment	Muscle	Motion	Sensation
Iliohypogastric	T ₁₂ -L ₁	-Internal oblique -External oblique -Transversus abdominus	Anterior abdominal wall	-Inferior abdominal wall -Upper lateral quadrant of buttock
Ilioinguinal	L ₁	Internal oblique	Anterior abdominal wall	-Inferior medial inguinal ligament -Genitalia
Genitofemoral	L ₁ -L ₂	Cremaster	Testicular	-Inferior medial inguinal ligament -Spermatic cord
Lateral femoral cutaneous	L ₂ -L ₃			-Anterior, lateral and posterior aspect of thigh
Femoral Nerve				
1. Anterior division	L ₂ -L ₄	-Sartorius -Pectineus	-Medial aspect of lower thigh -Adduction of thigh	-Anterior medial skin of thigh
2. Posterior division		Quadriceps	-Knee extension -Patellar movement	-Anterior thigh Hip and knee
Obturator Nerve				
1. Anterior division	L ₂ -L ₄	-Gracilis -Adductor longus -Adductor brevis -Pectineus	Thigh adduction	-Posterior medial thigh -Medial knee -Hip
2. Posterior division		-Adductor Magnus -Obturator externus	Thigh adduction with lateral hip rotation	-Knee

Relation between psoas & LB branches

□ Lateral border of psoas:

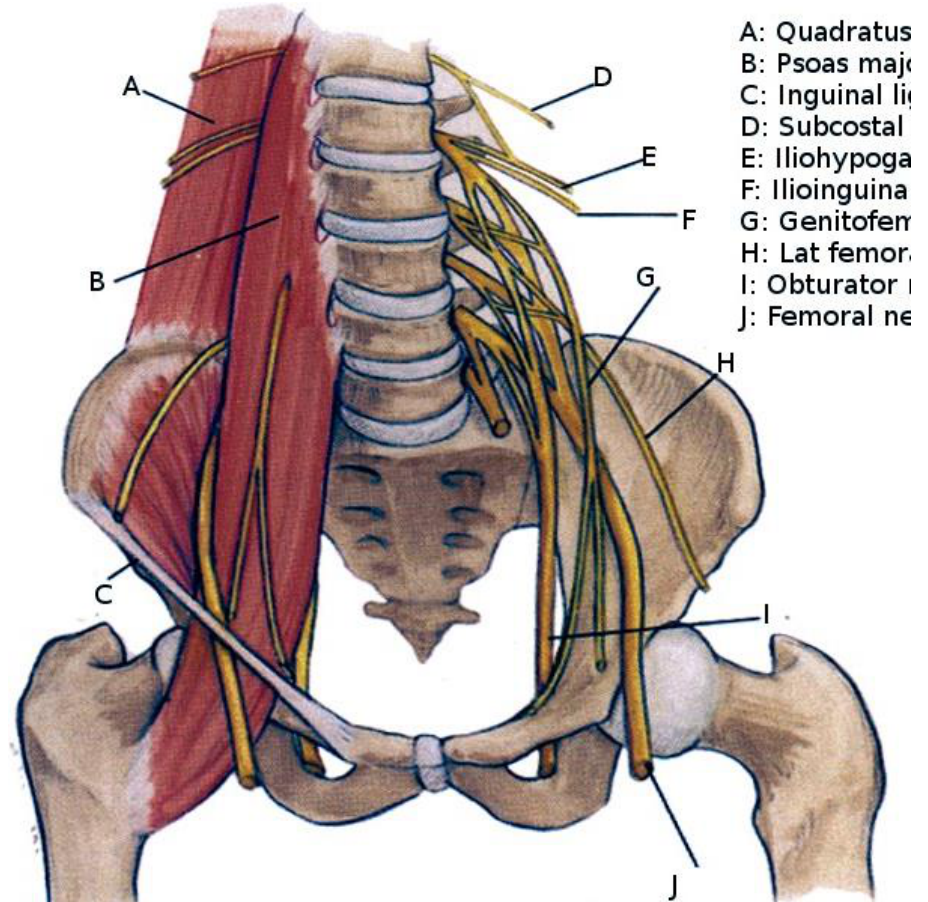
- Iliohypogastric & ilioinguinal nerves.
- Lateral femoral cutaneous nerve.

□ Anterior surface of psoas:

- Genitofemoral nerve.

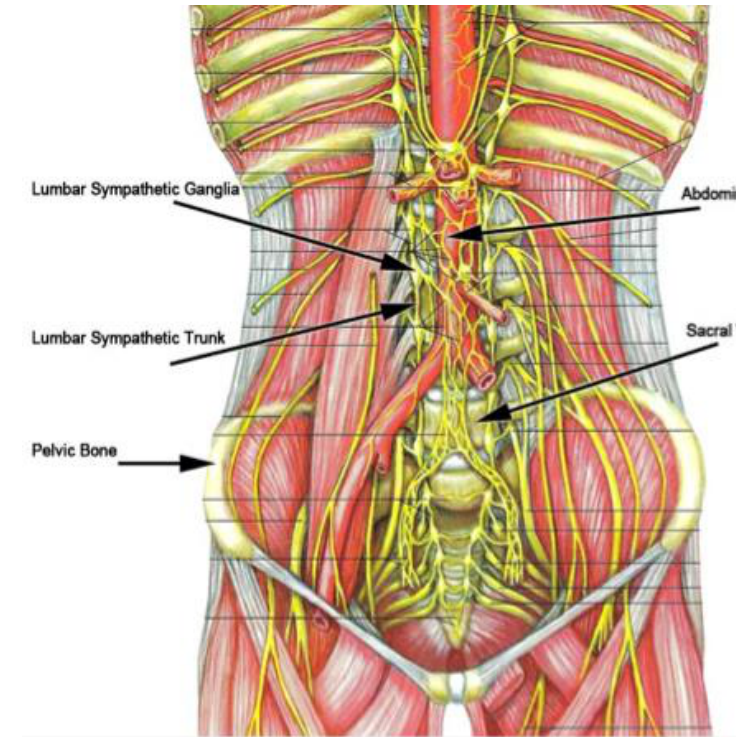
□ Medial border of psoas:

- Obturator nerve.



ANS in Abdomen

- Sympathetic & Parasympathetic.
- ❑ Sympathetic: derived from:
 - Lumbar part of sympathetic chain:
 - ✓ Gives both somatic and visceral branches:
 - ✓ Somatic branches supply lower abdominal wall & lower limb.
 - ✓ Visceral branches supply only pelvic organs.
 - Thoracic part of sympathetic chain through Celiac plexus:
 - ✓ Gives only visceral branches supplies all abdominal organs including gonads.
- ❑ Parasympathetic: from:
 - Vagus nerve.
 - Pelvic splanchnic nerve.



That is all

